WORKED SOLUTION

STEP II 1997, Question 11

Mechanics • Projectile Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

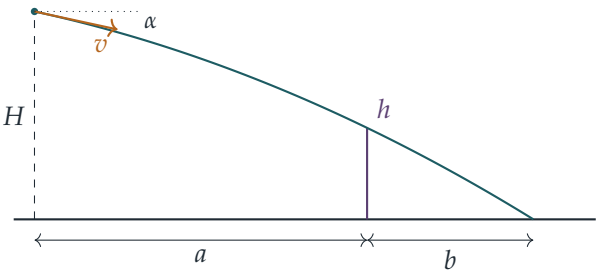
step.maths.org/questions/97-s2-q11

The key idea

Write down the trajectory as an equation for the drop below the server's height in terms of horizontal distance travelled. Then the two facts we are given (the ball is at height h when it has gone a, and at height 0 when it has gone $a+b$) are two equations in the two unknowns v and α. The bounds come from asking when the solutions of those equations actually make physical sense.

Solution

We model the ball as a particle moving under gravity alone (no air resistance or spin). Take the origin at the ground directly below the point where the ball is struck, with x horizontal towards the net and y vertically upwards.



The formula for v^2

Initially the velocity is $(v\cos\alpha, -v\sin\alpha)$. After time t,

$$x = vt\cos\alpha, \qquad y = H - vt\sin\alpha - \tfrac{1}{2}gt^2.$$

Eliminating $t = x/(v\cos\alpha)$ and using $\sec^2\alpha = 1+\tan^2\alpha$,

$$y = H - x\tan\alpha - \frac{gx^2(1+\tan^2\alpha)}{2v^2}. \qquad (*)$$

It is convenient to write $T = \tan\alpha$ and $k = \dfrac{g(1+T^2)}{2v^2}$, so that $y = H - xT - kx^2$.

The ball lands at $x = a + b$, so putting $y = 0$ there:

$$0 = H - (a+b)T - k(a+b)^2 \quad \Longrightarrow \quad k = \frac{H-(a+b)T}{(a+b)^2}. \tag{1}$$

Replacing k by its definition and rearranging for v^2 gives

$$v^2 = \frac{g(a+b)^2(1+\tan^2\alpha)}{2[H-(a+b)\tan\alpha]},$$

which is the first result.

The formula for $\tan\alpha$

The ball just clears the net, so $y = h$ at $x = a$. From $(*)$,

$$H - h = aT + ka^2.$$

Substituting k from (1) and multiplying through by $(a+b)^2$:

$$\begin{aligned}(H-h)(a+b)^2 &= aT(a+b)^2 + a^2\big[H-(a+b)T\big] \\ &= aT(a+b)\big[(a+b)-a\big] + a^2H \\ &= abT(a+b) + a^2H.\end{aligned}$$

Collecting the H terms, and using $(a+b)^2 - a^2 = b(2a+b)$,

$$ab(a+b)T = b(2a+b)H - (a+b)^2h.$$

Dividing by $ab(a+b)$, which is positive,

$$\tan\alpha = \frac{2a+b}{a(a+b)}\,H - \frac{a+b}{ab}\,h,$$

as required.

Bounds on H

Given a, b, h and H, the second formula fixes $\tan\alpha$ and then the first fixes v^2. Such a serve exists exactly when these values are admissible: $\tan\alpha > 0$ (the ball is hit *downwards*, so $0 < \alpha < \frac{\pi}{2}$) and $v^2 > 0$.

> **Ask yourself** The numerator of v^2 is clearly positive. So what does $v^2 > 0$ actually require?

It requires the denominator to be positive: $H - (a+b)\tan\alpha > 0$. Substituting the formula for $\tan\alpha$,

$$\begin{aligned}H - (a+b)\tan\alpha &= H - \frac{2a+b}{a}\,H + \frac{(a+b)^2}{ab}\,h \\ &= -\frac{a+b}{a}\,H + \frac{(a+b)^2}{ab}\,h = \frac{a+b}{a}\left[\frac{(a+b)h}{b} - H\right].\end{aligned}$$

Since $(a+b)/a > 0$, this is positive if and only if

$$H < \frac{(a+b)h}{b}.$$

For $\tan\alpha > 0$ we need

$$\frac{2a+b}{a(a+b)}H > \frac{a+b}{ab}h \quad\Longleftrightarrow\quad H > \frac{(a+b)^2 h}{b(2a+b)},$$

multiplying by the positive quantity $a(a+b)/(2a+b)$.

Both conditions together give

$$\boxed{\frac{(a+b)^2}{b(2a+b)}h \;<\; H \;<\; \frac{a+b}{b}h}$$

and conversely any H in this range gives a valid α and v. The interval is non-empty because $\dfrac{a+b}{2a+b} < 1$.

Both bounds have a clear meaning. The upper bound is the height at which the straight line from the landing point through the top of the net meets the server's vertical: gravity bends the path downwards, so the ball must start below that line (at $H = (a+b)h/b$ we would need infinite speed). The lower bound is the case $\alpha = 0$, a horizontal serve; serving from lower than this, the ball would have to be hit upwards to clear the net and still land at the given point.

Checks

- With $a = 12\,\text{m}$, $b = 6\,\text{m}$ and $h = 1\,\text{m}$ the bounds are $1.8\,\text{m} < H < 3\,\text{m}$. At $H = 1.8$ the formula gives $\tan\alpha = \frac{30}{216}(1.8) - \frac{18}{72} = 0$, as expected for the lower bound.
- Taking random a, b, h and H within the bounds, computing α and v from the two formulas and stepping the motion numerically, the ball passed through the top of the net and landed at $x = a+b$ every time (relative error below 10^{-14} for the exact path).
- Dimensions: each term of $\tan\alpha$ is a length divided by a length, and v^2 has dimensions of $g\times$ length, as it should.

Common slips

- Sign errors from the downward angle: the initial vertical velocity is $-v\sin\alpha$, and the $x\tan\alpha$ term in $(*)$ is subtracted.
- Stopping at "$v^2 > 0$" without noticing that it is a condition on the denominator, which only becomes a condition on H after substituting for $\tan\alpha$.

Beyond STEP

Here two boundary conditions (the net and the landing point) determine the launch conditions, rather than the other way round: a simple *boundary-value problem*, and the idea behind "shooting methods" for solving differential equations numerically. With air resistance and spin the equations no longer have closed-form solutions and this is exactly how one would proceed. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectiles with drag and much more.

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